

Fermi–Amaldi-Based Local Mixing Function

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(Dated: March 12, 2026)

I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

where

$$\gamma(\mathbf{r}, \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}') \quad (2)$$

is one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

Both quantities are strictly non-positive, hence, we will consider their non-negative absolute values, $|e_x^{\text{exact}}(\mathbf{r})|$ and $|e_x^{\text{FA}}(\mathbf{r})|$, for convenience.

For one-orbital systems (one-electron and two-electron singlet), two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_x^{\text{exact}}(\mathbf{r}) = e_x^{\text{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \quad (4)$$

II. SCRAP

A local hybrid exchange functional is given by

$$E_x^{\text{LH}} = \int \{g(\mathbf{r})e_x^{\text{exact}}(\mathbf{r}) + [1 - g(\mathbf{r})]e_x^{\text{SL}}(\mathbf{r})\} d\mathbf{r}, \quad (5)$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The e_x^{SL} and e_x^{exact} quantities are a semi-local and the exact exchange energy densities, respectively.

Here, we propose a new local mixing function for local hybrid functionals. This LMF is the ratio between the Fermi–Amaldi and the exact exchange energy densities:

$$g(\mathbf{r}) = \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}}. \quad (6)$$

Both exchange densities are manifestly non-positive and equivalent for one-electron systems and two-electron singlets. Although, one can show that

$$e_x^{\text{FA}} > e_x^{\text{exact}} \quad \text{when } N > 2 \quad (7)$$

by considering corresponding exchange holes. The exchange hole is defined via

$$E_x = \frac{1}{2} \int \int \frac{\rho(\mathbf{r})\rho_x(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad (8)$$

leading to expressions for the exact and the Fermi–Amaldi exchange holes:

$$\rho_x^{\text{exact}}(\mathbf{r}, \mathbf{r}') = -\frac{|\gamma(\mathbf{r}, \mathbf{r}')|^2}{2\rho(\mathbf{r})} \quad (9)$$

and

$$\rho_x^{\text{FA}}(\mathbf{r}') = -\frac{\rho(\mathbf{r}')}{N}. \quad (10)$$

Now, compare diagonal elements ($\mathbf{r} = \mathbf{r}'$) of the exact exchange hole to the Fermi–Amaldi exchange hole:

$$\rho_x^{\text{FA}} > \rho_x^{\text{exact}}|_{\mathbf{r}=\mathbf{r}'} \quad \text{when } N > 2. \quad (11)$$

Since, ρ_x^{exact} off-diagonal elements are negative, adding them to the diagonal part makes ρ_x^{exact} even more negative. Thus, for any number of electrons:

$$e_x^{\text{exact}} \leq e_x^{\text{FA}} \leq 0. \quad (12)$$

Combining these observations, we find that the proposed LMF is bound between zero and one:

$$0 \leq \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} \leq 1. \quad (13)$$

This inequality ensures that one hundred percent of exact exchange is employed in Eq. (5) for one-electron systems, satisfying the one-electron constraint [8, 9].

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (14)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

Another important constraint, the high-density limit,

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